

Luke Yarnall

Blacksburg VA, LukeYarn8836@gmail.com, **Mechanical Engineer** expected to graduate **Spring 2027**

Education

- **Virginia Polytechnic Institute and State University**, Blacksburg, VA (2024 - Expected Spring 2027)
 - Bachelor of Science in Mechanical Engineering (Current **GPA: 3.20**, Senior)
- **Northern Virginia Community College (NOVA)**, Annandale, VA (2022 - 2024)
 - Associate of Science in Engineering (**GPA: 3.682**, *Magna Cum Laude*).

Experience

- **Metal 3D Printing Manufacturing Intern MELD Printworks**, 415 National Dr, Christiansburg, VA 24073 - (2025 June - December, (second year) 2026 June-July) **Total: 7 months**
 - Supervised and maintained CNC-style aluminum 3D printers for aerospace and automotive components, ensuring proper calibration, environmental monitoring, quality control of materials, and safe shutdown procedures.
 - Conducted root cause analysis on print failures then have the plates milled and resurfaced.
 - Followed and documented Standard Operating Procedures (SOPs) for aluminum material and impurity removal, ensuring consistent material quality and compliance with manufacturing standards.
 - Customers include: Lockheed, Boeing, Hyundai, IAI, Sikorsky.
- **Printing and Manufacturing lead 3D-Printing Aircraft Competition Club (3D-PAC)**, 501 Industrial Park Rd, Blacksburg, VA 24060 (Sep 2024-current) **Total: 2 years**
 - Aerodynamic Design: Designed plane profiles and evaluated airframe aerodynamic stability using XFLR5 CFD software to optimize flight performance.
 - CAD Modeling: Modeled lightweight RC aircraft components and sub-assemblies using Fusion 360 and SolidWorks.
 - 3D Printing & Manufacturing: Manufactured aircraft using Ender 5 Pro, Bambu P1S, CR-30 (infinite-axis), and Creality K2 Pro 3D printers.
 - Printing & Manufacturing Lead: Repaired and maintained printing hardware, set production schedules, and managed print operations to build competition planes for events in Arlington, TX.
 - Competition Results: Earned **2nd place nationally in 2025** with MagicAirbus (14-second flight) and **2nd place in 2026** with HookieTail (22-second flight)

Skills

- **Additive & Manufacturing:** Metal Additive Manufacturing (MELD Friction Stir Process), FDM/SLA 3D Printing, CNC Machine Operation, Slicing (Bambu Studio, Cura, PrusaSlicer)
- **CAD & Prototyping:** SolidWorks, Fusion 360, NX Siemens, Onshape, Design for Manufacturability (DFM), Rapid Tooling Jigs & Fixtures
- **Quality & Analysis:** Root Cause Failure Analysis (RCA), Standard Operating Procedures (SOPs), Quality Control & Inspection, XFLR5 (CFD)
- **Programming & Tools:** Python, MATLAB, Java, Microcontrollers (Arduino), Analog transmitters and receivers (RC controllers and video transmitters)
- **Languages:** English (Native), Japanese (Fluent)

Projects

- **Atlas Program (I - VI):** Personal fully 3D printed RC plane design to fly, transmit video footage, and deliver payload (LinkedIn profile)
- **Swivel Sight:** Fully 3D printed accessory that stores firearm sight horizontally using bolt hinges and bungee cord.
- **Battle Cups:** In Course group project of an arduino that utilizes continuous servos, an ultrasonic sensor, and switches to carry cups around a table for a game of cup pong

Honors and Award

- **VA Governor's Academic Excellence Award** & Certificate of Recognition for Early College Scholars Program (2022), **VA Board of Education Biliteracy Award** (2022), **LBSS Wrestling Team Academic Excellence Letter** (2019-2022), NVCC Dean's List Honoree (Fall 2022), Nominated for the **National Society of Leadership and Success Award** (Spring 2023)